

Scenario-Based Dynamic Modeling Framework for Multi-Vector Hybrid Microgrids in Isolated Amazonian Defense Outposts

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Abstract

Comprehensive operational planning of hybrid microgrids in isolated equatorial environments, such as the Amazon rainforest, requires simulation tools capable of capturing environmental and logistical constraints across multiple energy vectors. Traditional deterministic approaches often obscure operational stress by relying on time-averaged assumptions. This paper presents a scenario-based dynamic modeling framework for multi-vector hybrid microgrids, integrating photovoltaic generation, lithium-ion battery storage, hydrogen-based seasonal storage, and a diesel backup generator under localized boundary conditions. A non-homogeneous Markov chain combined with solar geometry models is used to generate stochastic irradiance profiles, while a multi-layer demand model represents the volatile load of a remote Special Border Platoon. The framework is evaluated over an annual horizon using a rule-based dispatch strategy, revealing system-level stress patterns such as battery depletion events, hydrogen saturation, and fuel scarcity under constrained logistics. The source code is released as an open computational environment for microgrid design and control research. . The complete implementation is available at: <https://github.com/quantyma/quantyma-research/tree/main/quantyma-article-004>

Keywords: Hybrid Microgrids, Non-Homogeneous Markov Chains, Multi-Vector Energy Storage, Amazon Region, Energy-Logistics Simulation, Complex Systems, Dynamic Systems, Time-Series Analysis.

1 Introduction

The decarbonization and operational resilience of isolated infrastructure represent a major challenge in contemporary power systems engineering [Lasseter, 2002]. In equatorial regions such as the Amazon basin, remote military outposts known as Special Border Platoons (*Pelotão Especial de Fronteira*, PEF) operate under geographical isolation and rely heavily on diesel fuel supply chains constrained by seasonal river navigation. The operational costs, environmental impacts, and logistical dependency associated with continuous diesel transport motivate the deployment of hybrid renewable microgrids to improve local energy autonomy [Rezaei et al., 2021].

However, microgrid sizing and multi-vector energy management for isolated tropical applications cannot rely exclusively on deterministic tools or simplified monthly meteorological averages [Sinha and Chandel, 2014]. The Amazon region presents strong short-term solar intermittency driven by cloud convection and rapidly changing atmospheric conditions. In parallel, military operational schedules produce irregular and time-varying electrical demand profiles.

The integration of multi-vector storage architectures, combining short-term Battery Energy Storage Systems (BESS) with long-duration Hydrogen (H_2) storage, introduces coupled dynamics across multiple temporal scales that directly affect system reliability [Parra et al., 2019]. Under variable renewable generation and constrained fuel logistics, conventional heuristic dispatch strategies may fail to anticipate long-duration resource depletion events, increasing the probability of unmet load conditions during extended low-water periods.

To address these challenges, this paper presents a stochastic time-series simulation framework designed to capture coupled energy and logistics interactions in isolated Amazonian microgrids. The proposed architecture integrates dynamic models for photovoltaic generation, electrochemical and hydrogen storage systems, electrical demand variability, and diesel fuel inventory constraints within a unified simulation environment. A Rule-Based Controller (RBC) is adopted as a baseline dispatch strategy to evaluate operational bottlenecks and system-level vulnerabilities over long-horizon simulations. The

resulting framework establishes a reproducible computational benchmark for future research involving predictive and learning-based energy management methods.

2 Modeling Dynamic Solar Irradiance, Electric Demand and Microgrid System

This section describes the mathematical structure and operational constraints of the proposed simulation framework for isolated hybrid microgrids. Conventional deterministic approaches and time-averaged representations often fail to capture relevant variability in energy and logistics dynamics under constrained environments.

To address this limitation, the framework is organized into three coupled components: (i) stochastic boundary conditions for solar irradiance and demand profiles, (ii) state-space models for energy storage systems and generation assets, and (iii) a rule-based dispatch strategy reflecting operational priorities. These components are integrated into a unified discrete-time simulation environment that enables consistent evaluation of system behavior under scenario-driven conditions.

2.0.1 Dynamic Solar Irradiance Modeling

To establish realistic boundary conditions for the hybrid microgrid simulation without relying on incomplete or oversimplified regional telemetry, a hierarchical stochastic-geometric framework was developed to model the hourly horizontal solar irradiance $G(t)$ (W/m^2) for Manaus, Brazil (Latitude -3.11°).

2.0.1.1 Seasonal Dynamics and Daily Weather State Transitions: The macro-climatic behavior shifts continuously between wet and dry seasons. A time-dependent seasonal factor $SF(d)$ is defined for each day of the year $d \in [1, 365]$, anchoring the peak wet period around February/March ($SF \rightarrow 0$) and the peak dry period around August/September ($SF \rightarrow 1$):

$$SF(d) = 0.5 \cdot \left(1 - \cos \left(2\pi \frac{d - 110}{365} \right) \right) \quad (1)$$

The daily weather state $S_d \in \{\text{sunny, partial, rainy}\}$ establishes persistent meteorological conditions across a 24-hour block. The transition probability matrix from day $d-1$ to day d is dynamically updated by interpolating between two empirical boundary matrices (T_{wet} and T_{dry}) using the seasonal factor:

$$P_{\text{interp}}(d) = (1 - SF(d)) \cdot T_{\text{wet}}[S_{d-1}] + SF(d) \cdot T_{\text{dry}}[S_{d-1}] \quad (2)$$

where the steady-state transition matrices are parameterized as:

$$T_{\text{wet}} = \begin{bmatrix} 0.25 & 0.45 & 0.30 \\ 0.10 & 0.50 & 0.40 \\ 0.05 & 0.25 & 0.70 \end{bmatrix}, \quad T_{\text{dry}} = \begin{bmatrix} 0.85 & 0.12 & 0.03 \\ 0.45 & 0.45 & 0.10 \\ 0.30 & 0.40 & 0.30 \end{bmatrix} \quad (3)$$

The daily state S_d is selected via a random choice driven by $P_{\text{interp}}(d)$ and broadcasted across all 24 hours of the respective day. Each state maps to an hourly atmospheric attenuation multiplier $\omega_{\text{weather}}(t)$, where $\omega_{\text{sunny}} = 1.0$, $\omega_{\text{partial}} = 0.50$, and $\omega_{\text{rainy}} = 0.20$.

2.0.1.2 Deterministic Geometric Envelope: The maximum theoretical clear-sky irradiance ceiling varies seasonally ($G_{\text{max,base}}$) and incorporates a daily normal disturbance factor ($\mathcal{N}(1.0, 0.05)$) to simulate day-to-day atmospheric variability:

$$G_{\text{max,dynamic}}(d) = [680 + (G_{\text{max,dry}} - 680) \cdot SF(d)] \cdot \mathcal{N}(1.0, 0.05) \quad (4)$$

where $G_{\text{max,dry}}$ is set to $1000 \text{ W}/\text{m}^2$. The instantaneous clear-sky profile $G_{\text{clear}}(t)$ is governed by the solar zenith angle (θ_z), tracking the solar declination (δ) and the local hour angle (H) for hour t :

$$\cos(\theta_z) = \sin(\phi) \sin(\delta) + \cos(\phi) \cos(\delta) \cos(H) \quad (5)$$

$$G_{\text{clear}}(t) = \max(0, \cos(\theta_z)) \cdot G_{\text{max,dynamic}}(d) \quad (6)$$

2.0.1.3 Stochastic High-Frequency Disturbances: The deterministic weather-attenuated profile ($G_{\text{clear}}(t) \cdot \omega_{\text{weather}}(t)$) is combined with an additive Gaussian noise profile to emulate high-frequency cloud turbulence. The standard deviation of this noise is scaled by an interpolated state-dependent multiplier (M_{noise}):

$$M_{\text{noise}}(t) = (1 - SF(d)) \cdot M_{\text{wet}}[S_d] + SF(d) \cdot M_{\text{dry}}[S_d] \quad (7)$$

where $M_{\text{wet}} = [0.2, 0.6, 0.1]$ and $M_{\text{dry}} = [0.05, 0.3, 0.15]$ for sunny, partial, and rainy states, respectively. The final computed irradiance $G(t)$ enforces strict physical caps and handles night/twilight cuts using a threshold parameter ($G_{\text{threshold}} = 45.0 \text{ W/m}^2$):

$$G_{\text{noisy}}(t) = G_{\text{clear}}(t) \cdot \omega_{\text{weather}}(t) + \mathcal{N}(0, (M_{\text{noise}}(t) \cdot \sigma_{\text{noise}})^2) \cdot \mathbb{I}_{\{\cos(\theta_z) > 0\}} \quad (8)$$

$$G(t) = \begin{cases} 0.0, & \text{if } \cos(\theta_z) \leq 0 \text{ or } (G_{\text{clear}}(t) \cdot \omega_{\text{weather}}(t)) < G_{\text{threshold}} \\ \max(0.0, \min(1350.0, G_{\text{noisy}}(t))), & \text{otherwise} \end{cases} \quad (9)$$

where $\sigma_{\text{noise}} = 300$ and $\mathbb{I}$ is the indicator function isolating solar activity to astronomical daytime.

2.0.2 Tactical Electrical Demand Modeling

To capture the volatile, multi-layered operational profile of an isolated Special Border Platoon (*Pelotão Especial de Fronteira* - PEF), the total instantaneous electrical power demand $P_{\text{demand}}(t)$ (W) is modeled as a multi-vector composition of persistent defensive infrastructure, stochastic operational missions, and dynamic thermal-cooling mechanisms.

2.0.2.1 Elevated Baseline Infrastructure Load: The critical surveillance radar, tactical communications, and continuous perimeter security arrays operate uninterrupted (24/7). This baseline power consumption $P_{\text{base}}(t)$ is tied directly to the macro-climatic season factor $SF(t)$, capturing increased infrastructure stress during dry regimes:

$$P_{\text{base}}(t) = [0.35 + (0.42 - 0.35) \cdot SF(t)] \cdot P_{\text{peak}} \quad (10)$$

where P_{peak} represents the calibrated theoretical maximum peak power of the entire military facility (50 000 W).

2.0.2.2 Stochastic Active Operational Load: Human regimentary schedules and tactical mission activities are mapped using a customized Gaussian Mixture Model (GMM). The daily activity profiles differentiate between standard workdays ($\mathcal{P}_{\text{workday}}$) and weekend regimes ($\mathcal{P}_{\text{weekend}}$) across hour $h \in [0, 23]$:

$$\mathcal{P}_{\text{workday}}(h) = 0.65 \cdot \exp\left(-\left(\frac{h-10}{2.5}\right)^2\right) + 0.55 \cdot \exp\left(-\left(\frac{h-15}{2.0}\right)^2\right) + 0.05 \quad (11)$$

$$\mathcal{P}_{\text{weekend}}(h) = 0.35 \cdot \exp\left(-\left(\frac{h-11}{3.0}\right)^2\right) + 0.40 \cdot \exp\left(-\left(\frac{h-16}{2.5}\right)^2\right) + 0.05 \quad (12)$$

To implement realistic volatility, the deterministic operational profile $\mathcal{P}(t) \in \{\mathcal{P}_{\text{workday}}, \mathcal{P}_{\text{weekend}}\}$ is modulated by a dual-scale controllable stochastic engine. Inter-day variance (structural shifts per day d) and intra-day variance (high-frequency hourly chattering) are integrated via independent normal distributions, yielding:

$$\eta_{\text{day}}(d) \sim \mathcal{N}(1.0, \sigma_{\text{daily}}^2), \quad \eta_{\text{hour}}(t) \sim \mathcal{N}(1.0, \sigma_{\text{hourly}}^2) \quad (13)$$

$$P_{\text{operational}}(t) = \max(0, \mathcal{P}(t) \cdot (0.4 \cdot P_{\text{peak}}) \cdot \eta_{\text{day}}(d) \cdot \eta_{\text{hour}}(t)) \quad (14)$$

where the inter-day volatility index is set to $\sigma_{\text{daily}} = 0.25$ and the hourly noise is $\sigma_{\text{hourly}} = 0.15$.

2.0.2.3 Dynamic Thermal and Cooling Load Architecture: The refrigeration demand for defense server centers and munitions depots is highly dependent on continuous ambient heat accumulation. This behavior is modeled via a first-order continuous-discrete autoregressive thermal network. The raw thermal stress $E_{\text{thermal}}(t)$ captures the immediate solar irradiance gain ($\alpha_{\text{gain}} = 0.00042$) and the structure's passive ventilation heat loss rate ($\beta_{\text{loss}} = 0.22$), scaled by a seasonal dynamic factor:

$$\lambda_{\text{thermal}}(t) = 0.90 + (1.10 - 0.90) \cdot SF(t) \quad (15)$$

$$E_{\text{thermal}}(t) = [E_{\text{thermal}}(t-1) \cdot (1 - \beta_{\text{loss}}) + \alpha_{\text{gain}} \cdot G(t)] \cdot \lambda_{\text{thermal}}(t) \quad (16)$$

The raw energy series is normalized and mapped to match the target maximum cooling capacity envelope, allocating up to 30% of the facility's total energy budget to thermal management:

$$P_{\text{thermal}}(t) = \left(\frac{E_{\text{thermal}}(t)}{\max_k \{E_{\text{thermal}}(k)\}} \right) \cdot (0.30 \cdot P_{\text{peak}}) \quad (17)$$

2.0.2.4 Aggregate Compilation and Lower Bound Constraints: High-frequency grid surges and secondary auxiliary subsystems are introduced through an unmodulated additive white Gaussian noise vector $\epsilon(t) \sim \mathcal{N}(0, (0.05 \cdot P_{\text{peak}})^2)$. The preliminary tactical demand is compiled as the sum of all active layers. Finally, to represent the strict survival and continuous defense posture of the base, a lower regressive boundary floor is enforced, guaranteeing that the power draw never drops below 30% of total capacity:

$$P_{\text{demand, raw}}(t) = P_{\text{base}}(t) + P_{\text{operational}}(t) + P_{\text{thermal}}(t) + \epsilon(t) \quad (18)$$

$$P_{\text{demand}}(t) = \max(0.30 \cdot P_{\text{peak}}, P_{\text{demand, raw}}(t)) \quad (19)$$

2.1 Microgrid Asset Subsystem Modeling

To evaluate the dynamic interactions within the hybrid architecture, each energy vector is mathematically modeled based on its discrete-time operational constraints, power boundaries, and state transition efficiencies under a simulation step of $\Delta t = 1$ h.

2.1.1 Photovoltaic Generation Model

The instantaneous electrical power produced by the photovoltaic array $P_{\text{solar}}(t)$ (W) is calculated as a direct linear transformation of the incident horizontal solar irradiance $G(t)$, the conversion efficiency of the modules η_{PV} , and the aggregate effective surface area:

$$P_{\text{solar}}(t) = G(t) \cdot (N_{\text{panels}} \cdot A_{\text{panel}}) \cdot \eta_{\text{PV}} \quad (20)$$

where N_{panels} is the total number of physical panels, and A_{panel} represents the unit surface area (m^2).

2.1.2 Multi-Vector Electrochemical and Chemical Storage Modeling

The short-term Battery Energy Storage System (BESS) and the long-term Hydrogen Storage System (H_2) are formulated using localized state-space capacity models. Let index $k \in \{\text{BESS}, H_2\}$ denote the respective storage vector. Each subsystem is characterized by its nominal energy capacity $E_{\text{cap},k}$ (Wh), maximum power rating $P_{\text{max},k}$ (W), round-trip charging efficiency $\eta_{\text{ch},k}$, discharging efficiency $\eta_{\text{dis},k}$, and a lower operational safety boundary $E_{\text{min},k}$ (Wh).

The active power request dispatched to the unit is strictly bound by its physical power rating:

$$P_{\text{req},k}(t) = \max(-P_{\text{max},k}, \min(P_{\text{dispatched},k}(t), P_{\text{max},k})) \quad (21)$$

where $P_{\text{req},k} > 0$ denotes a charging command, and $P_{\text{req},k} < 0$ denotes a discharging command. The dynamic tracking of the stored energy state $E_k(t)$ and its corresponding State of Charge (SoC_k) follows a piecewise conditional logic:

2.1.2.1 Case I: Charging Process ($P_{\text{req},k} > 0$)

$$E_k(t) = \begin{cases} E_k(t-1) + P_{\text{req},k}(t) \cdot \eta_{\text{ch},k} \cdot \Delta t, & \text{if } E_k(t-1) + P_{\text{req},k}(t) \cdot \eta_{\text{ch},k} \cdot \Delta t \leq E_{\text{cap},k} \\ E_{\text{cap},k}, & \text{otherwise} \end{cases} \quad (22)$$

$$P_{\text{actual},k}(t) = \begin{cases} P_{\text{req},k}(t), & \text{if capacity is not saturated} \\ \frac{E_{\text{cap},k} - E_k(t-1)}{\eta_{\text{ch},k} \cdot \Delta t}, & \text{at saturation limit} \end{cases} \quad (23)$$

2.1.2.2 Case II: Discharging Process ($P_{\text{req},k} < 0$)

$$E_k(t) = \begin{cases} E_k(t-1) - \frac{|P_{\text{req},k}(t)|}{\eta_{\text{dis},k}} \cdot \Delta t, & \text{if } E_k(t-1) - \frac{|P_{\text{req},k}(t)|}{\eta_{\text{dis},k}} \cdot \Delta t \geq E_{\text{min},k} \\ E_{\text{min},k}, & \text{otherwise} \end{cases} \quad (24)$$

$$P_{\text{actual},k}(t) = \begin{cases} P_{\text{req},k}(t), & \text{if above minimum boundary} \\ -\left(\frac{E_k(t-1) - E_{\text{min},k}}{\Delta t}\right) \cdot \eta_{\text{dis},k}, & \text{at depletion limit} \end{cases} \quad (25)$$

The normalized state of charge percentage is computed at each time step as:

$$\text{SoC}_k(t) = \left(\frac{E_k(t)}{E_{\text{cap},k}}\right) \cdot 100.0 \quad (26)$$

The parameterized boundary conditions embedded within the simulation engine are defined as:

- **BESS Architecture:** $\eta_{\text{ch}, \text{BESS}} = 0.92$, $\eta_{\text{dis}, \text{BESS}} = 0.92$, and $E_{\text{min}, \text{BESS}} = 0.10 \cdot E_{\text{cap}, \text{BESS}}$.
- **Hydrogen Architecture:** $\eta_{\text{ch}, H_2} = 0.65$ (electrolyzer efficiency), $\eta_{\text{dis}, H_2} = 0.55$ (fuel cell stack efficiency), and $E_{\text{min}, H_2} = 0.05 \cdot E_{\text{cap}, H_2}$.

2.1.3 Backup Diesel Generator Model

The backup internal combustion diesel generator acts as the final system security vector, constrained by its maximum active capacity $P_{\text{gen}, \text{max}}$ (W). Given a net power deficit requirement $P_{\text{needed}}(t)$, the power reference commanded to the generator is capped by:

$$P_{\text{ref}}(t) = \min(P_{\text{needed}}(t), P_{\text{gen}, \text{max}}) \quad (27)$$

The hourly fuel consumption profile $\dot{V}_{\text{fuel}}(t)$ (L) depends linearly on the active power generated via the specific consumption parameter α_{fuel} (L/Wh), where $\alpha_{\text{fuel}} = \frac{\text{consumption rate [L/kWh]}}{1000}$.

The dynamic inventory depletion of the tactical fuel stock $V_{\text{tank}}(t)$ (L) handles finite fuel availability according to the remaining volume:

$$V_{\text{tank}}(t) = \begin{cases} V_{\text{tank}}(t-1) - \dot{V}_{\text{fuel}}(t), & \text{if } V_{\text{tank}}(t-1) \geq P_{\text{ref}}(t) \cdot \alpha_{\text{fuel}} \cdot \Delta t \\ 0.0, & \text{otherwise} \end{cases} \quad (28)$$

$$P_{\text{diesel}}(t) = \begin{cases} P_{\text{ref}}(t), & \text{if fuel is available} \\ \frac{V_{\text{tank}}(t-1)}{\alpha_{\text{fuel}} \cdot \Delta t}, & \text{under partial fuel-starvation limits} \end{cases} \quad (29)$$

where any reduction below $P_{\text{ref}}(t)$ due to total fuel depletion ($V_{\text{tank}} \rightarrow 0$) forces a dispatch mismatch, triggering immediate system vulnerability tracking.

The system control is governed by a deterministic cascade heuristic. At each time step t , the controller evaluates $P_{\text{net}}(t)$ and executes the following priority order:

1. Condition I ($P_{\text{net}} \geq 0$): Surplus Management

- Allocate surplus power to charge the BESS until $\text{SoC}_{\text{BESS}, \text{max}}$ is achieved.
- Redirect remaining excess power to the H_2 electrolyzer up to $P_{\text{elec}, \text{max}}$.
- Curtail any remaining power if both storage systems are fully saturated.

2. Condition II ($P_{\text{net}} < 0$): Deficit Management

- Discharge the BESS to meet the deficit until reaching $\text{SoC}_{\text{BESS},\min}$ or hitting $P_{\text{BESS},\max}$.
- If a deficit persists, activate the H_2 Fuel Cell up to $P_{\text{FC},\max}$.
- If the storage assets are depleted, ignite the backup Diesel Generator, tracking the fuel extraction from $V_{\text{tank}}(t)$.

3. Condition III: System Breakdown

- If $V_{\text{tank}}(t) = 0$ or the load step exceeds the maximum power limits of the operating units, the remaining deficit is registered as Unmet Load ($P_{\text{unmet}}(t)$), indicating a microgrid blackout condition.

2.2 System Parameterization and Simulation Setup

Table 1: Calibrated Microgrid Simulation Parameters and Subsystem Boundary Conditions.

Subsystem Vector	Operational Parameter	Value	Unit
General Operational	Peak Power Demand (P_{peak})	50000.0	W
	Total Simulation Horizon	5	Years
	Annual Simulation Steps (n_{steps})	8760	Hours
Photovoltaic Array	Number of Panels (N_{panels})	42	–
	Unit Panel Area (A_{panel})	2.0	m ²
	Conversion Efficiency (η_{PV})	0.22	–
Battery Storage (BESS)	Nominal Energy Capacity ($E_{\text{cap, BESS}}$)	60000.0	Wh
	Maximum Active Power ($P_{\text{max, BESS}}$)	10000.0	W
	Initial State of Charge ($\text{SoC}_{\text{BESS}, 0}$)	0.80	–
	Charging Efficiency ($\eta_{\text{ch, BESS}}$)	0.92	–
	Discharging Efficiency ($\eta_{\text{dis, BESS}}$)	0.92	–
	Lower Safety Boundary ($E_{\text{min, BESS}}$)	6000.0	Wh
Hydrogen Storage	Tank Energy Capacity ($E_{\text{cap, } H_2}$)	100000.0	Wh
	Maximum System Power ($P_{\text{max, } H_2}$)	5000.0	W
	Initial Tank Inventory ($\text{SoC}_{H_2, 0}$)	0.50	–
	Electrolyzer Efficiency ($\eta_{\text{ch, } H_2}$)	0.65	–
	Fuel Cell Efficiency ($\eta_{\text{dis, } H_2}$)	0.55	–
	Lower Safety Boundary ($E_{\text{min, } H_2}$)	5000.0	Wh
Backup Diesel Gen.	Rated Generation Capacity ($P_{\text{gen, max}}$)	10000.0	W
	Fuel Tank Capacity (V_{full})	500.0	L
	Specific Fuel Consumption (α_{fuel})	0.25	L/kWh
	Logistic Refueling Threshold ($V_{\text{threshold}}$)	50.0	L

3 Results and Discussion

To evaluate the storage dynamics and dispatch behavior produced by the baseline Rule-Based Controller (RBC), a granular time-series inspection of the simulated multi-vector microgrid operation is presented in Figure 1. The controller operates through a deterministic priority-based cascade logic, where instantaneous photovoltaic generation is first compared against the active electrical demand to compute the net power balance (P_{net}). Under surplus conditions ($P_{\text{net}} > 0$), the available excess renewable energy is sequentially allocated to the Battery Energy Storage System (BESS) and subsequently to the Hydrogen (H_2) subsystem, respecting the maximum power limits and operational constraints of each asset.

Conversely, during deficit conditions ($P_{\text{net}} < 0$), the controller prioritizes energy extraction from the BESS as the primary short-term balancing mechanism. If the deficit persists, the hydrogen fuel-cell subsystem is activated as a secondary energy vector, followed by the diesel generator as the final backup source when renewable and storage resources become insufficient. This simplified dispatch strategy intentionally avoids predictive or optimization-based behavior, allowing the simulations to expose the intrinsic operational limitations of reactive rule-based energy management under constrained environmental and logistical conditions.

The operational analysis is organized across three domains: primary power balances, active asset dispatch trajectories, and internal capacity inventory states.

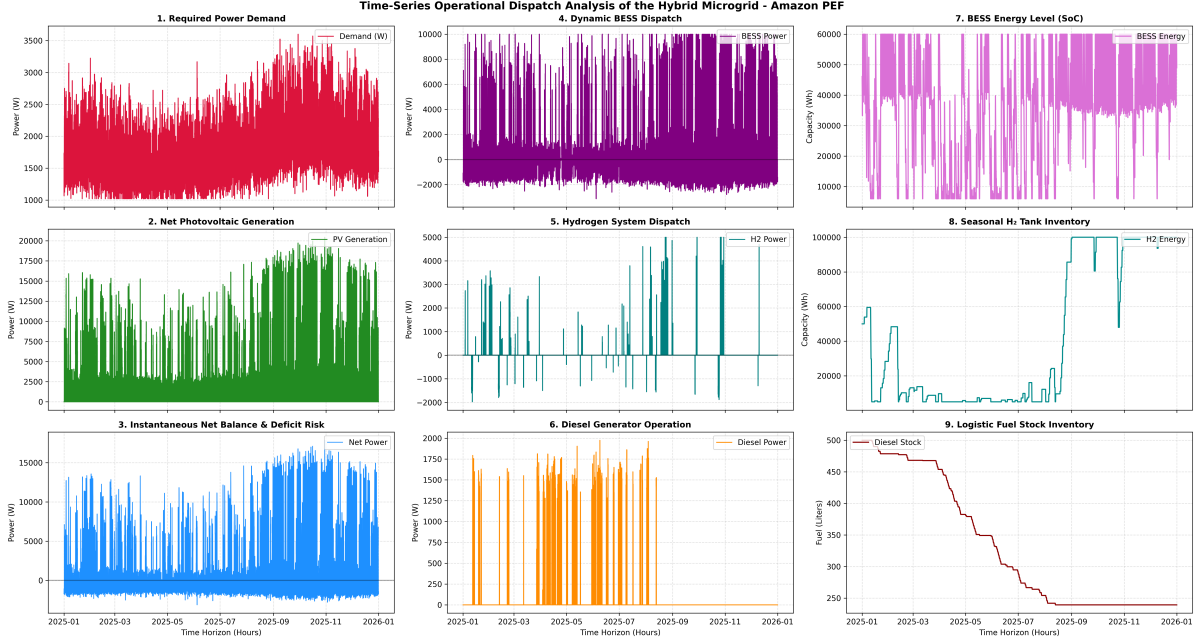


Figure 1: Time-series operational analysis of the simulated hybrid microgrid environment. Column 1 presents the primary power balance variables, including power demand, photovoltaic generation, and instantaneous net balance. Column 2 shows the active dispatch behavior of the BESS, hydrogen storage system, and diesel generator under the Rule-Based Controller (RBC). Column 3 illustrates internal storage states, operational boundaries, and unmet load events (P_{unmet}) observed during constrained operating conditions.

3.0.1 Primary Power Balances and Deficit Thresholds

As shown in Column 1 of Figure 1, the system equilibrium is governed by high-variability demand and renewable generation profiles. The *Required Power Demand* (Plot 1) exhibits volatile load behavior, with elevated demand periods concentrated between August and November and peak values approaching approximately 3500 W. Concurrently, the *Net Photovoltaic Generation* (Plot 2) represents the primary renewable energy source of the microgrid, displaying strong daily intermittency combined with a higher seasonal baseline during dry-season conditions.

The resulting *Instantaneous Net Balance* (Plot 3) computes the difference between instantaneous generation and demand ($P_{\text{net}} = P_{\text{PV}} - P_{\text{demand}}$). Positive excursions indicate periods of surplus renewable generation available for storage charging, whereas negative excursions represent deficit conditions associated with potential unmet load events ($P_{\text{unmet}} > 0$).

3.0.2 Active Asset Dispatch Trajectories

Column 2 isolates the active power contributions delivered by the energy management assets. The *Dynamic BESS Dispatch* (Plot 4) demonstrates high-frequency bidirectional operation, with the battery system acting as the primary short-term dynamic buffer. Charging and discharging cycles are continuously modulated to compensate for short-term solar ramp fluctuations and demand variability.

In contrast, the *Hydrogen System Dispatch* (Plot 5) exhibits smoother seasonal-scale behavior. The electrolyzer absorbs excess renewable generation during surplus conditions, while the fuel cell dispatches energy during prolonged renewable generation shortfalls. This coordinated behavior reduces dependence on the *Diesel Generator Operation* (Plot 6), whose utilization is primarily concentrated during the first half of the simulated horizon and progressively decreases during high-irradiance periods.

3.0.3 Capacity States and Logistical Inventories

The internal storage and inventory trajectories shown in Column 3 illustrate the long-term operational behavior of the simulated system. The *BESS Energy Level* (Plot 7) continuously modulates within its operational limits, indicating sustained daily cycling without violating the minimum state-of-charge constraint.

The *Seasonal H_2 Tank Inventory* (Plot 8) characterizes the long-term storage dynamics of the hydrogen subsystem. During the first half of the annual horizon, the hydrogen inventory remains near low-capacity operating conditions due to limited renewable surplus. Entering the high-irradiance season, sustained excess renewable generation progressively increases the stored hydrogen inventory, eventually approaching the maximum storage capacity.

This seasonal transition is also reflected in the *Logistic Fuel Stock Inventory* (Plot 9), where diesel fuel consumption decreases steadily throughout the simulation horizon. During periods of elevated renewable generation availability, diesel utilization approaches near-zero operation, indicating reduced dependence on fossil-fuel backup generation under the simulated conditions.

4 Conclusion and Future Work

The proposed simulation framework successfully computed a continuous annual operating horizon, integrating multi-vector energy dynamics into a unified time-series environment. The generated scenarios enabled the analysis of operational stress conditions that are often not represented by deterministic or time-averaged approaches.

The scenario-based irradiance and demand profiles produced recurring short-term battery depletion events, forcing the BESS to approach its minimum operational boundary ($E_{\min, \text{BESS}} = 6000.0 \text{ Wh}$) during periods of elevated variability. Seasonal fluctuations also produced extended hydrogen storage saturation periods, highlighting long-term balancing limitations within the simulated architecture.

The simulations further illustrated the interaction between energy dispatch and fuel logistics constraints. Under delayed fuel resupply conditions associated with low-water river transport scenarios, the baseline Rule-Based Controller (RBC) depleted both short- and long-term storage reserves without anticipatory conservation behavior. Once the diesel fuel inventory reached depletion ($V_{\text{tank}} \rightarrow 0$), residual demand deficits resulted in unmet load events ($P_{\text{unmet}} > 0$), representing localized system blackout conditions within the simulated environment.

Future work will utilize the proposed framework as an open computational environment for evaluating advanced energy management strategies, including Model Predictive Control (MPC) and Deep Reinforcement Learning (DRL) approaches for multi-vector scheduling under constrained logistics scenarios.

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